# Python recreation of Kuznetsov et al. (1994)

A 2-ODE model of tumor-immune dynamics, fit to $BCL_1$ lymphoma data in chimeric mice.

Note: figures 3, 4, and 5 are the most important.

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# Background

## Biological Motivation

$BCL_1$ lymphoma in chimeric mice shows both tumor dormancy and escape. Why do some immune systems contain tumors indefinitely while others fail? Kuznetsov et al. (1994) build a minimal ODE model to answer this.

## Original Model

Two populations: effector immune cells (E) and tumor cells (T). After quasi-steady-state reduction of conjugate dynamics, the system collapses to two equations:

$$\frac{dE}{dt} = s + \frac{pET}{g+T} - mET - dE$$
$$\frac{dT}{dt} = aT(1-bT) - nET$$

# Model Dissected

Saturating recruitment

Source Interaction Death

$$\frac{dE}{dt} = s + \frac{pET}{g+T} - mET - dE$$

$$\frac{dT}{dt} = aT(1 - bT) - nET$$

Logistic growth Interaction

Tumor grows logistically; effectors are recruited by tumor but also inactivated by it.

# Nondimensionalized Model

By making the following substitutions, we get a nondimensional model with 7 parameters.

$$x = \frac{E}{E_0} \qquad \eta = \frac{g}{T_0}$$

$$y = \frac{T}{T_0} \qquad \mu = \frac{m}{n} = \frac{k_3}{k_2}$$

$$\tau = nT_0 t \qquad \delta = \frac{d}{nT_0}$$

$$\sigma = \frac{s}{nE_0T_0} \qquad \alpha = \frac{a}{nT_0}$$

$$\rho = \frac{p}{nT_0} \qquad \beta = bT_0$$

where $E_0$, $T_0$ = 10^6 cells.

These two will be varied in Figure 4 in a bifurcation diagram

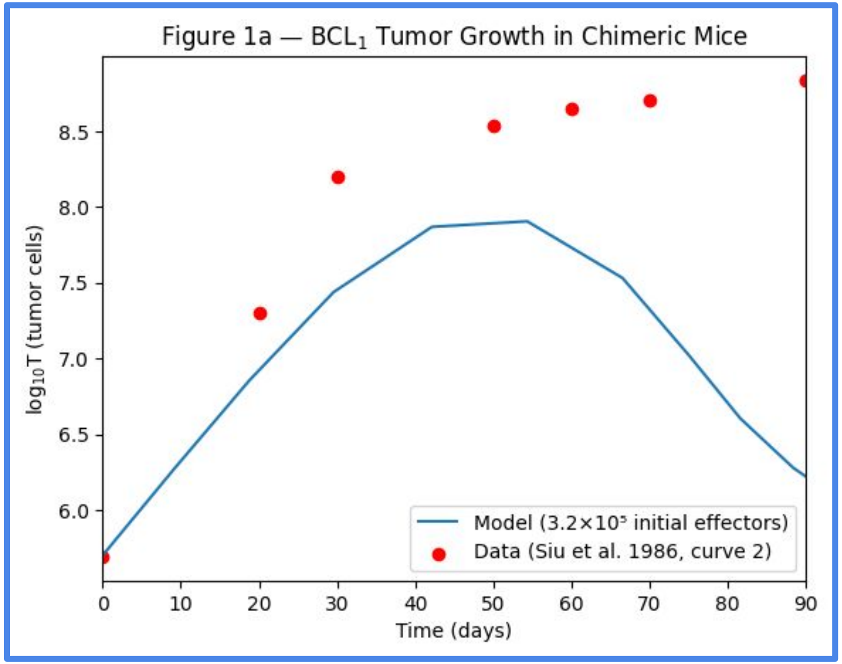
$$\frac{dx}{d\tau} = \sigma + \frac{\rho x y}{\eta + y} - \mu x y - \delta x$$

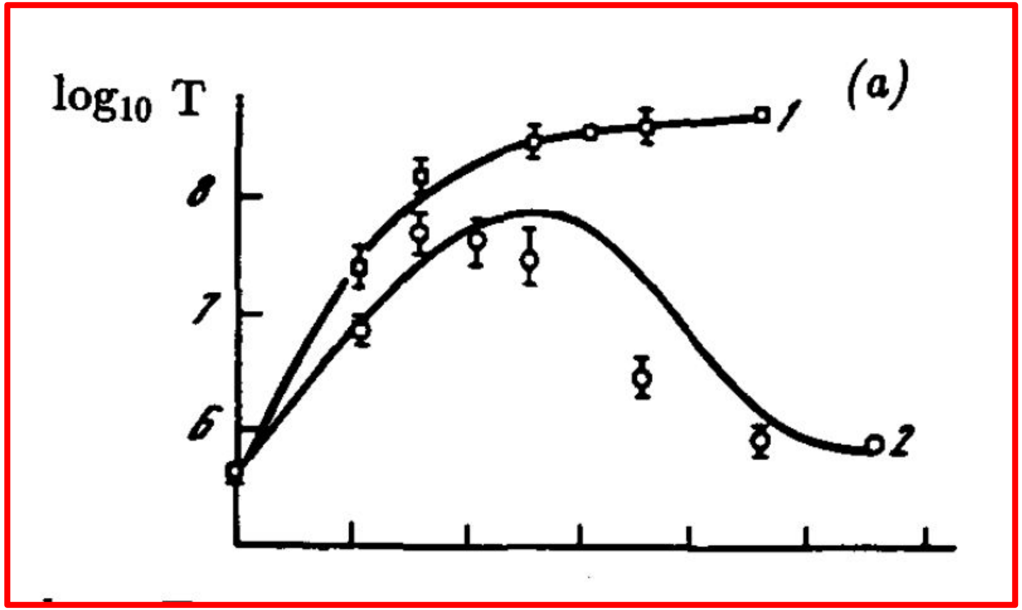
$$\frac{dy}{d\tau} = \alpha y(1 - \beta y) - xy$$

The critical parameter is μ, the ratio of effector inactivation to tumor kill rate. It governs whether sneaking-through is possible. It is the focus of figures 6, 8, and 9.

# Figure 1

Model fit to experimental $BCL_1$ tumor growth data. The immune system initially suppresses the tumor before it escapes. Model undershoots the plateau slightly.



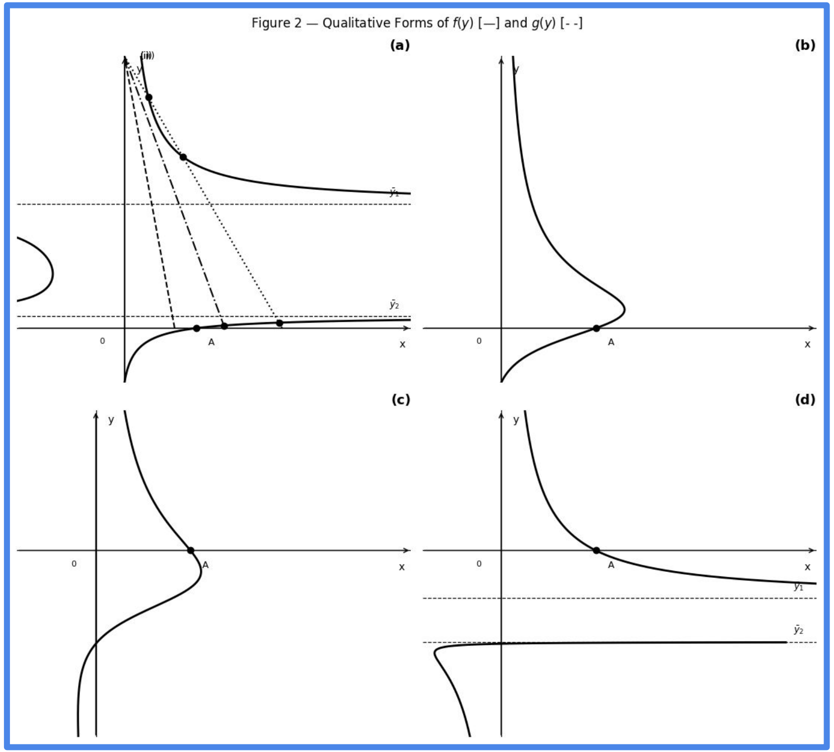



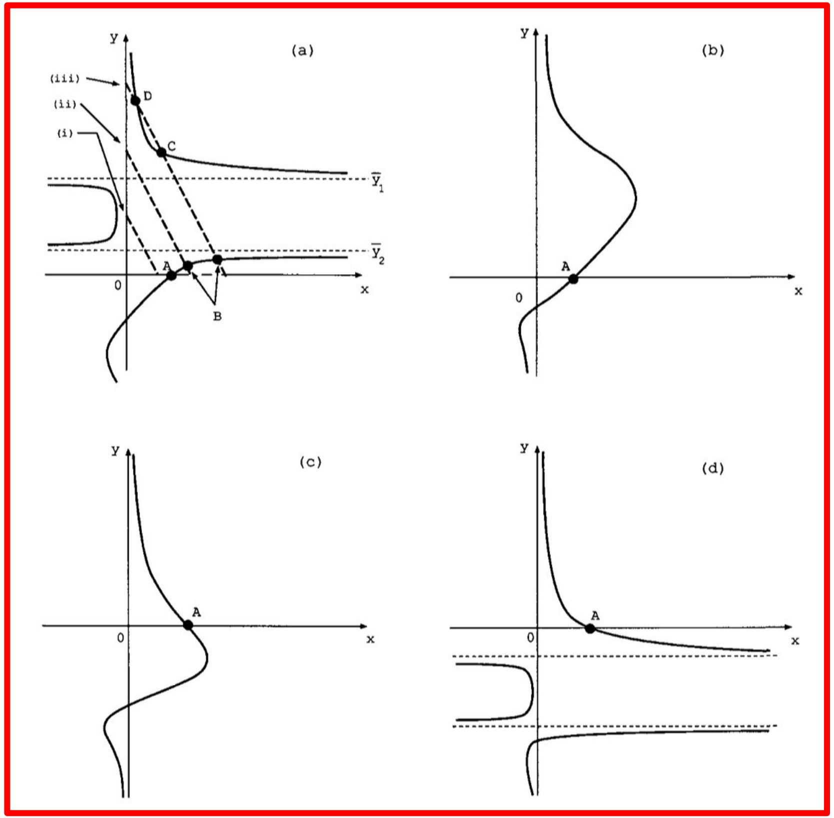

My figures will be outlined in blue

The paper's will be outlined in red

# Figure 2 (schematic)

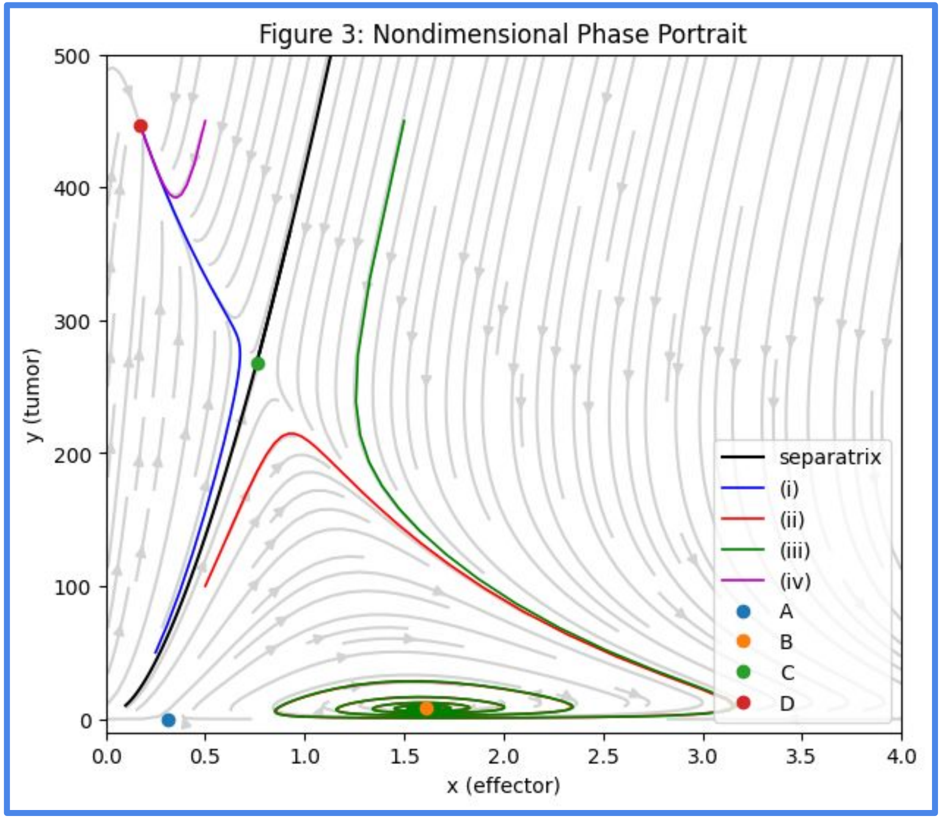
How the shape of the x-nullcline f(y) changes with recruitment parameter ρ. The number of intersections with g(y) determines how many steady states exist.

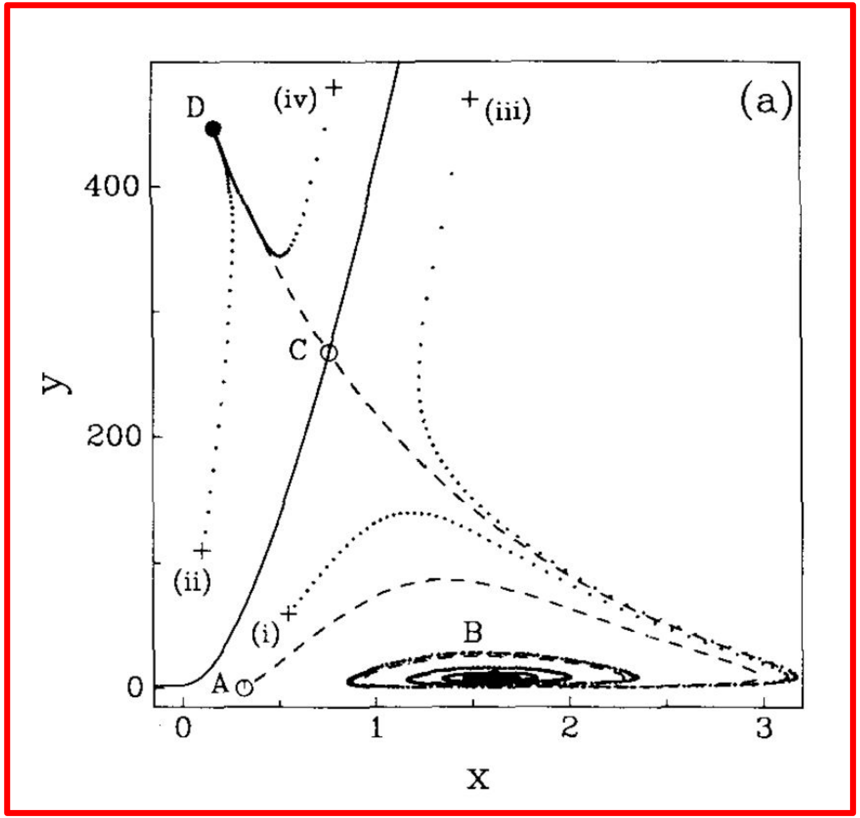


# Figure 3

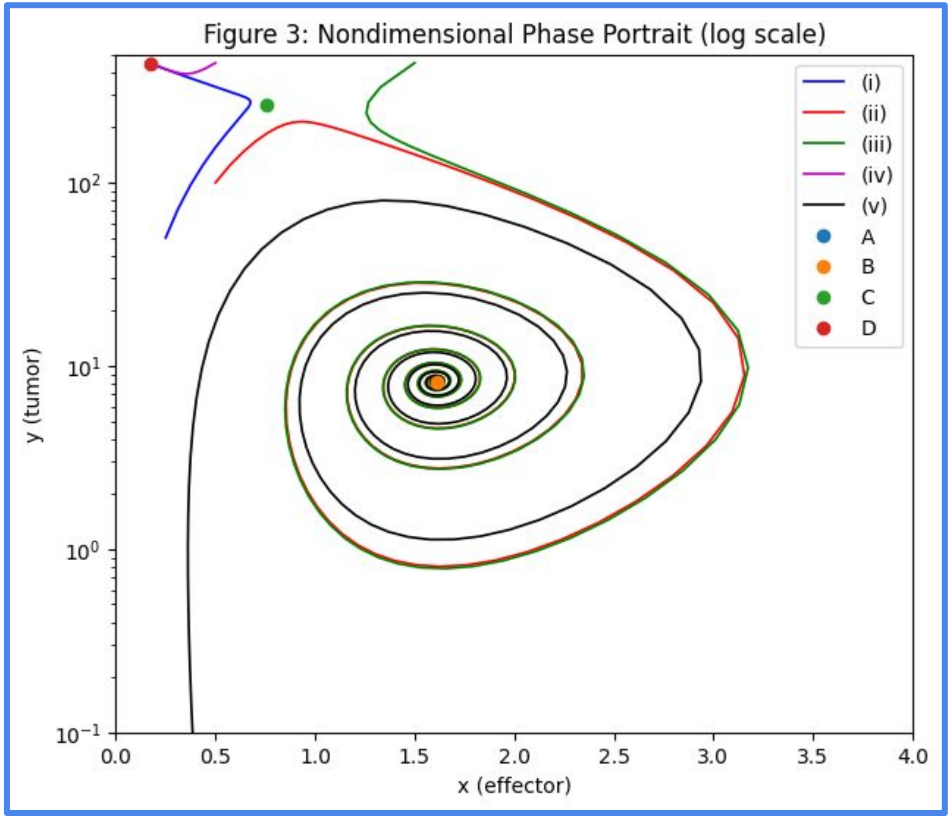
Phase portrait at estimated parameters. The separatrix (stable manifold of saddle C) divides the plane: below it leads to dormancy (B), above it leads to tumor escape (D).

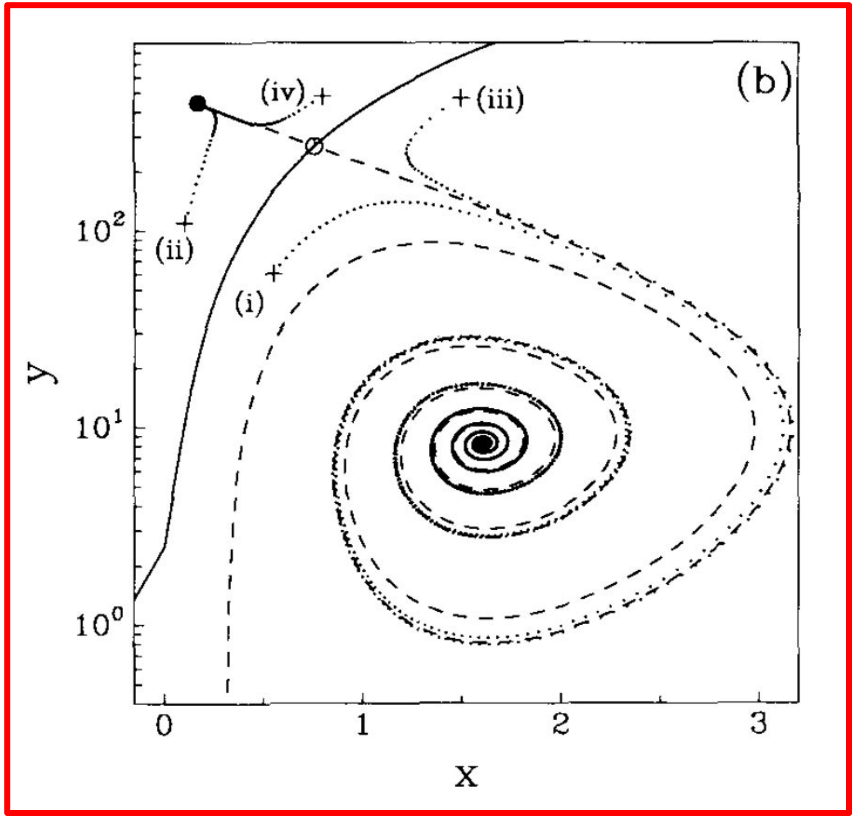


# Figure 3 (log)

Previous figure with a log y axis, which better displays the spiral into (B).

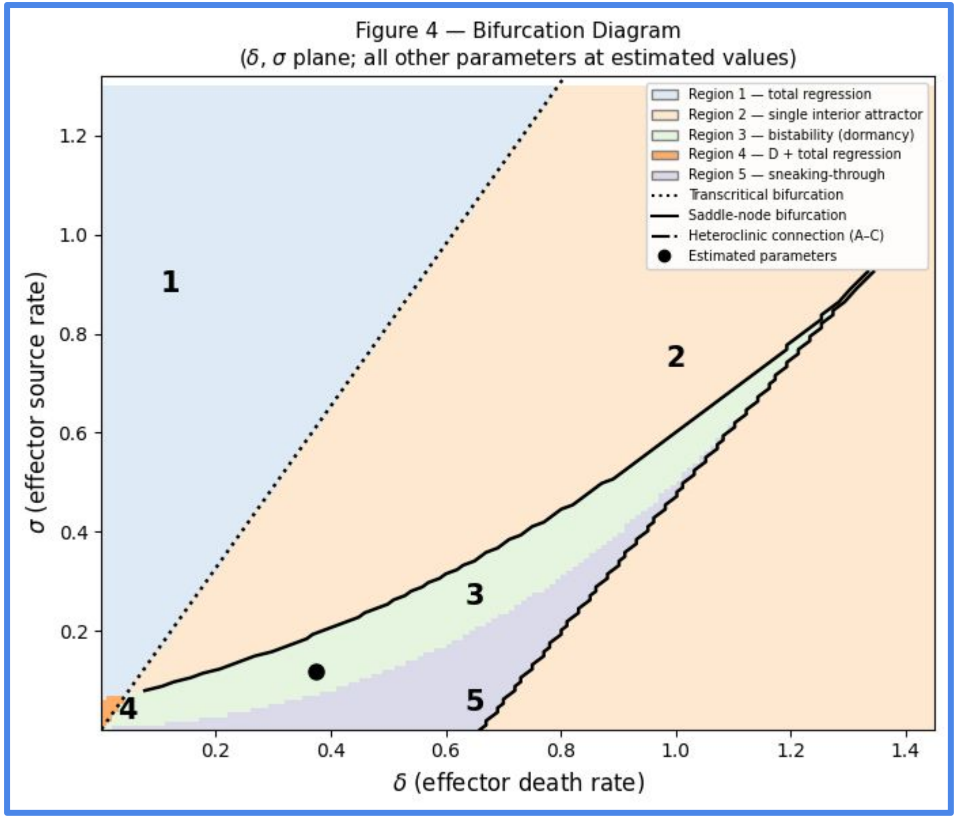


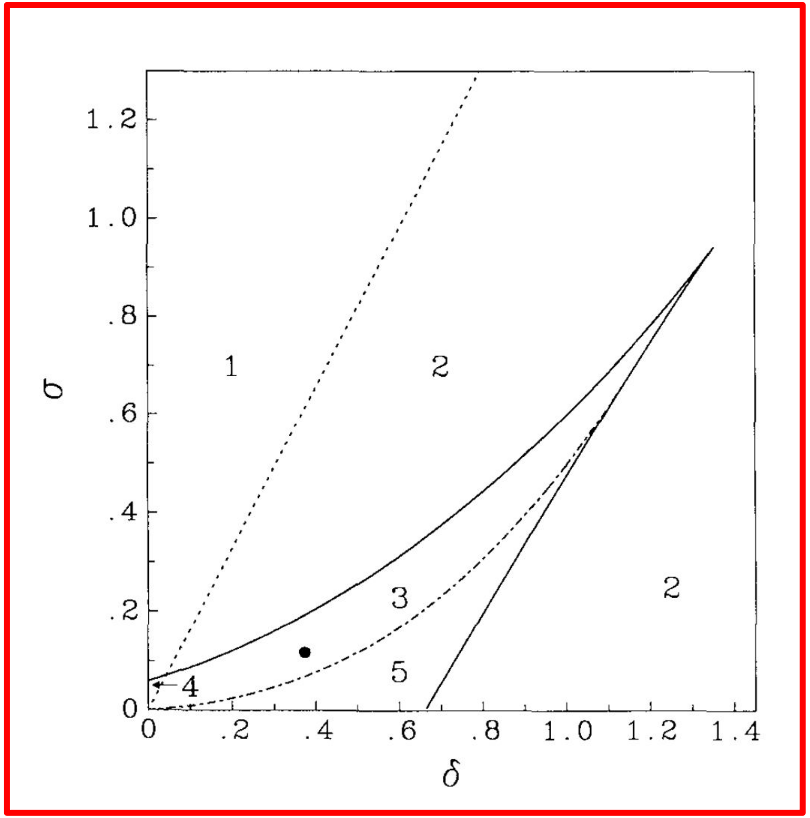



# Figure 4

Bifurcation diagram in the (δ, σ) plane. Five dynamical regions. Estimated parameters sit in Region 3, just above the heteroclinic boundary with Region 5

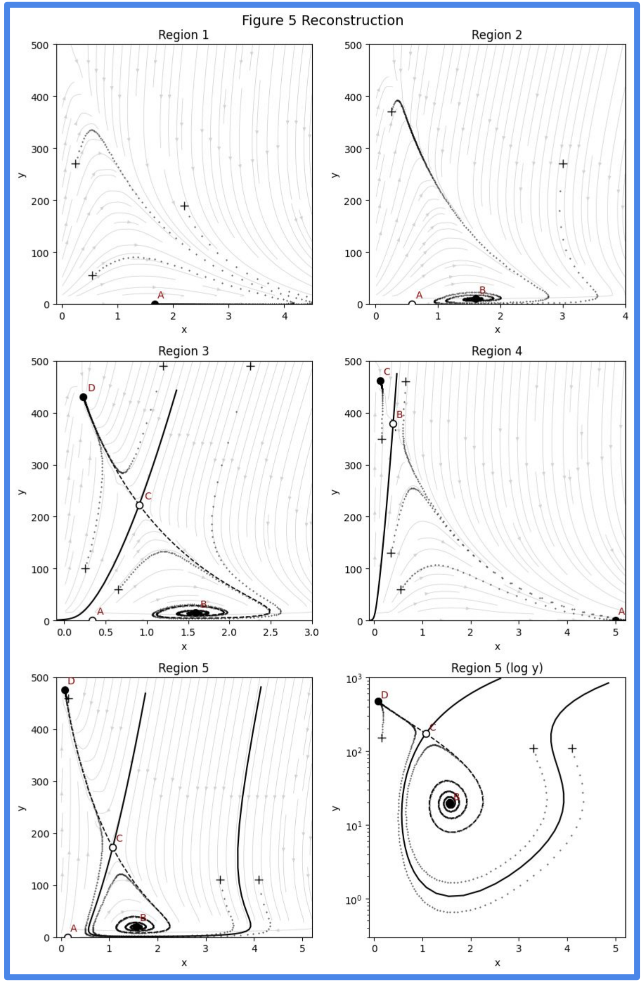
Transitions:
- 1-2, 4-3 : transcritical bifurcation (A and B)
- 1-4, 2-3 : saddle-node bifurcation (C and D)
- 2-5 : saddle-node bifurcation (C and B)
- 3-5 : heteroclinic connection (A and C)

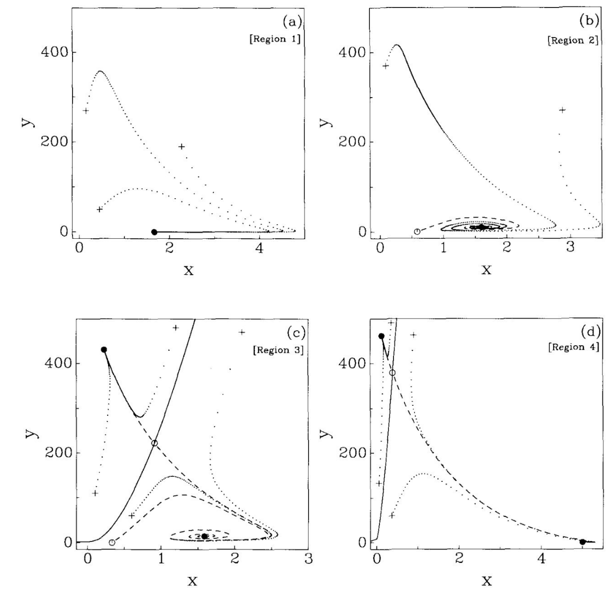


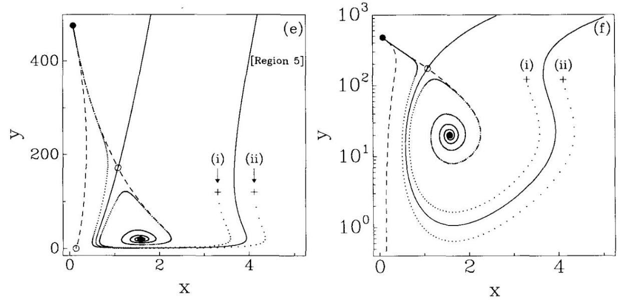
# Figure 5

One phase portrait per region. Regions 1–4 have local bifurcations. Region 5 (sneaking-through) requires detecting the global heteroclinic connection. 5 is reproduced on log y axis to better showcase the sneaking-through.



Figures 5a–d.



Figures 5e and f.

# Figure 6 (Schematic)

As μ crosses ~0.005014, the heteroclinic connection between A and C shifts the basin boundary dramatically, enabling sneaking-through.

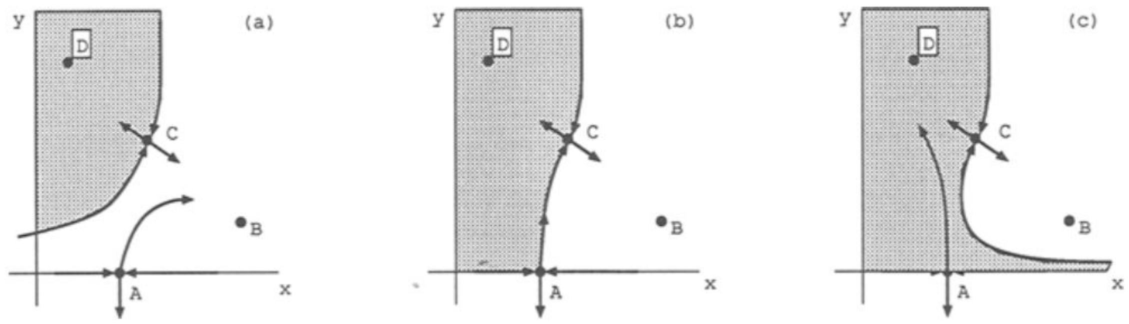
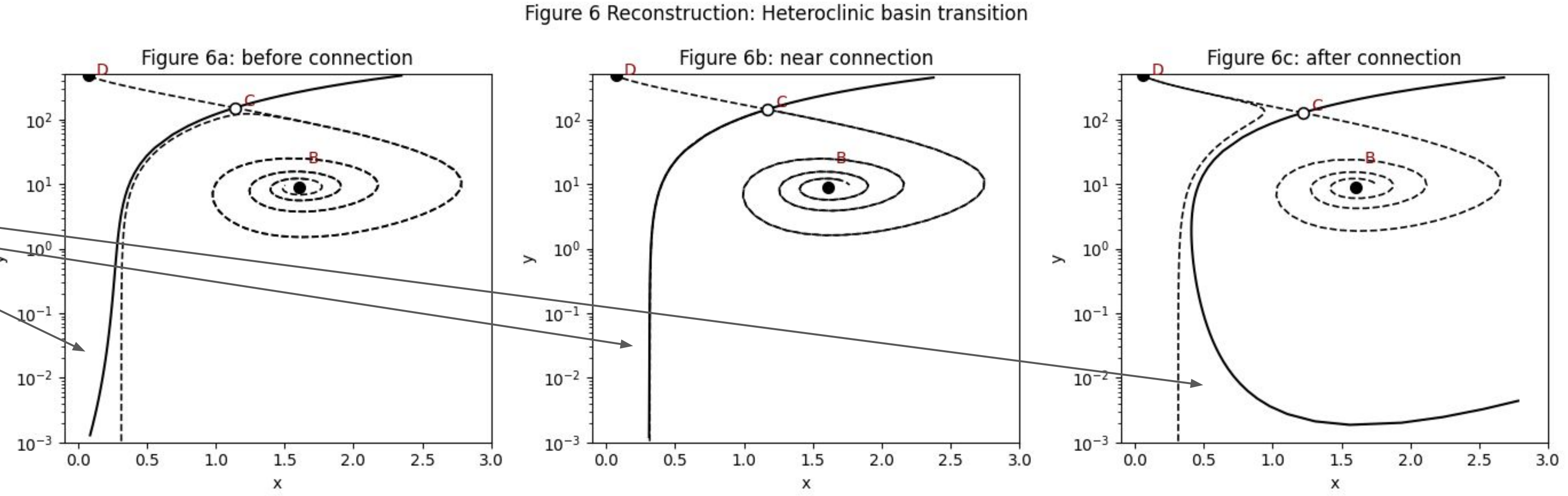


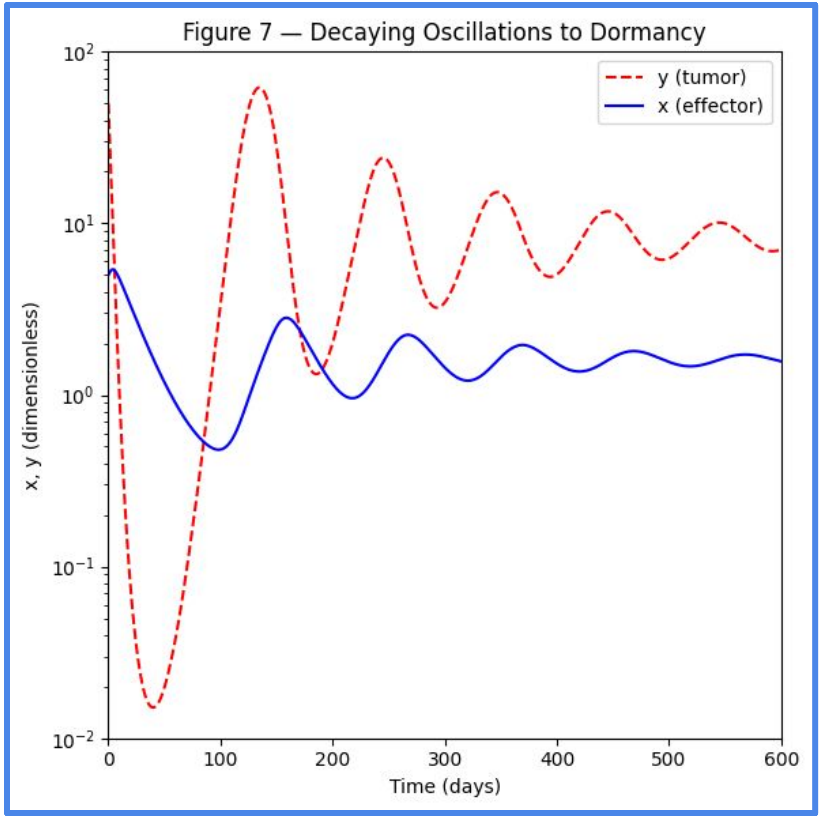

Figure 6. Representative phase portraits: (a) before, (b) during, and (c) after; the heteroclinic connection between steady states $A$ and $C$. Portions of the basins of attraction for attractors $B$ and $D$ are the white and shaded regions, respectively.

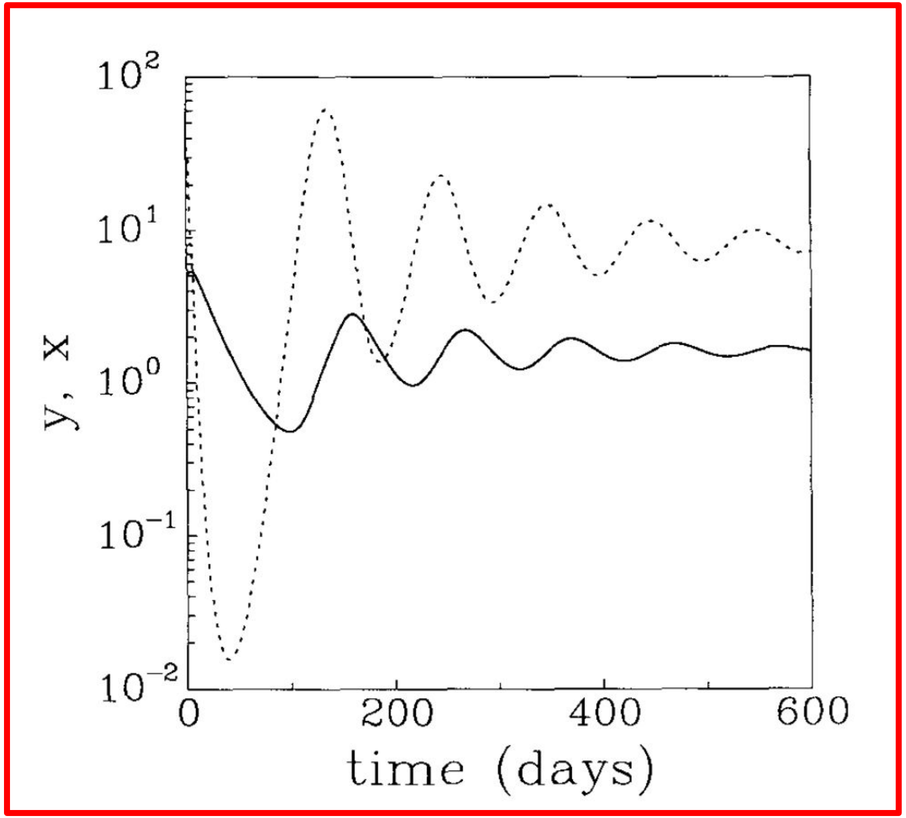
Notice the behavior of the tail



# Figure 7

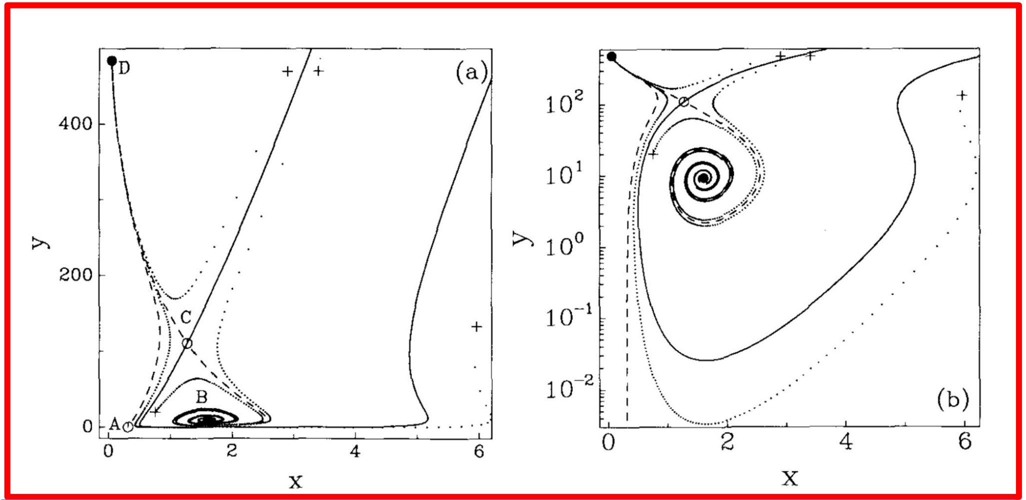
Time series from $(x_0, y_0) = (5, 50)$. Decaying oscillations converge to dormancy over ~600 days, with a ~3–4 month oscillation period.

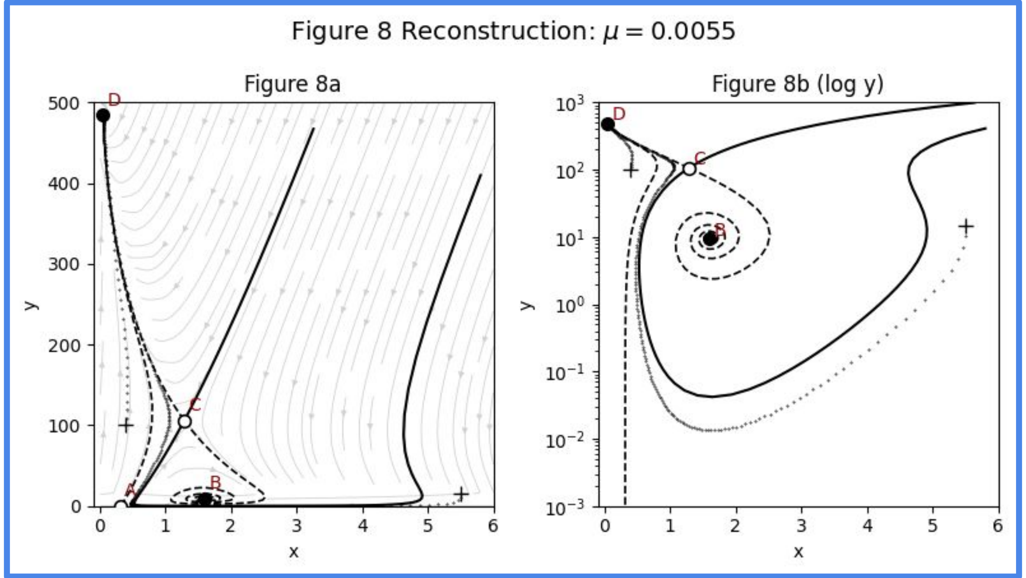


# Figure 8

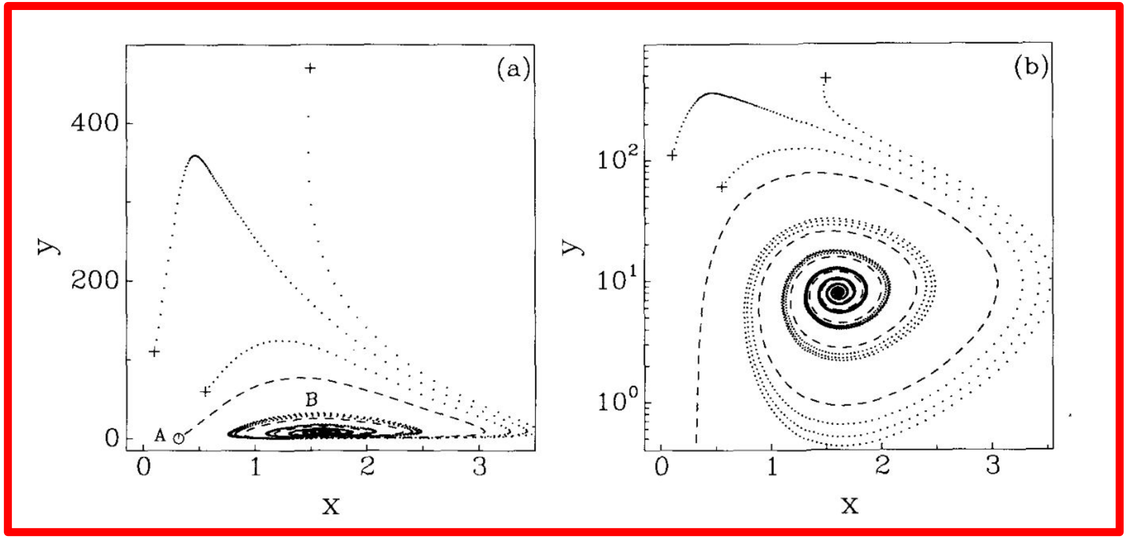
Past the heteroclinic bifurcation, the system is in Region 5. Small tumors can now sneak past immune defenses and escape

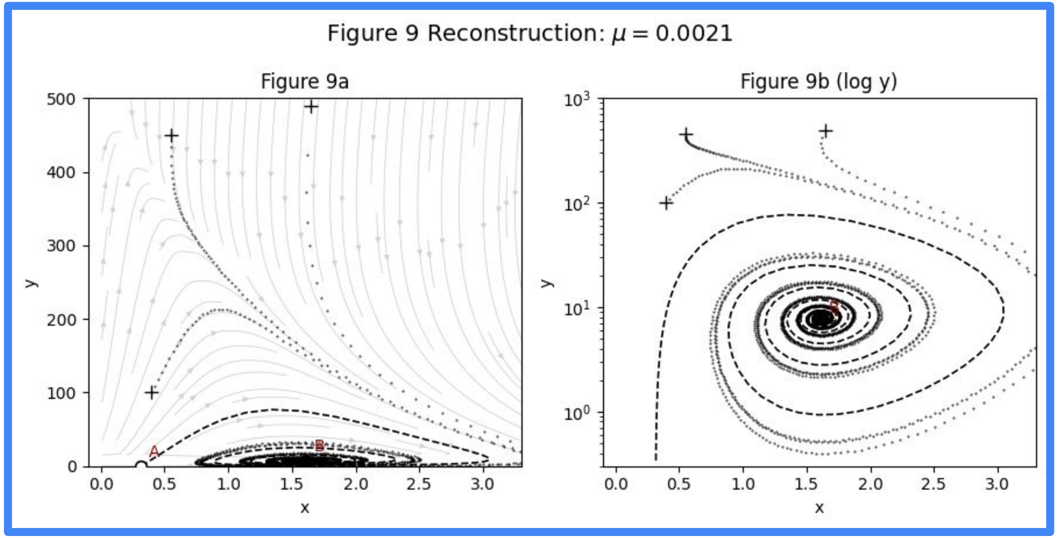


# Figure 9

At μ = 0.0021 (past the saddle-node bifurcation), C and D have annihilated. Only dormancy remains, and  tumor escape is no longer possible.



# Conclusion

The model reproduces tumor dormancy, escape, and sneaking-through from just two ODEs. Again, the key parameter is μ, the ratio of effector inactivation to tumor kill rate. The estimated value sits between two bifurcation thresholds, placing $BCL_1$ near a 'vital barrier' where small biological fluctuations can tip the system from dormancy into uncontrolled growth. This predicts that tumor dormancy is ultimately unstable, and that measuring a tumor's ability to inactivate effector cells is a meaningful clinical signal.